

ASK TWICE, LOOK TWICE: PROMPT ECHOING RESOLVES THE QUESTION-FIRST PARADOX IN VISION-LANGUAGE MODELS

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ABSTRACT

Whether a question is placed before or after an image changes nothing about a prompt’s content, yet it changes a vision-language model’s accuracy by up to 18 points. Question-first prompting, the layout that lets every image token attend to the question, costs Qwen3-VL-8B 7.0 points on average across 13 benchmarks and drives LLaVA-1.5 to exactly zero group accuracy. Degraded perception does not explain this: question-first makes $2.49\times$ *more* image patches decode a content word from the question, and it measurably rewrites the image tokens themselves. The ordering that looks most like it found the answer is the one that answers wrong. We locate the failure in read-out. At the peak read-out layer, question-first sends $2.19\times$ less attention from the answer position to the question, the two orderings are indistinguishable at chance through layer 25 of 36, and they then diverge to $P(\text{correct})$ of 0.0986 versus 0.8747. Attention knockout makes the dependence causal and doubly dissociated: severing answer-to-question costs question-last 5.6 points and question-first nothing, while severing answer-to-image costs question-first 9.6 points and question-last nothing. The diagnosis prescribes the fix. Echoing the question after the image costs 13 tokens and recovers 78% of the deficit while provably leaving perception untouched; echoing the image recovers all of it and adds 1.18 points over question-last. The echo works by position, not content: a spatially reversed echo is statistically indistinguishable from a correct one, and an eighth-resolution echo carrying $\frac{1}{64}$ of the pixels matches a full one at 3% of the added tokens.

1 INTRODUCTION

Ask a vision-language model a question about an image and you must decide, often without thinking about it, whether the question goes before the image or after it. The information content is identical either way. Under causal attention, putting the question first is the option that lets every image token see it, which is what a reader would want when told to look for something specific.

It makes the model worse. Moving the question before the image costs Qwen3-VL-8B 8.1 points of NaturalBench group accuracy ($p = 1.3 \times 10^{-12}$), 9.5 points on Winoground, and 18.0 points on CV-Bench, with an average loss of 7.0 points across 13 benchmarks. The same reversal holds for Qwen2.5-VL-7B, InternVL3-8B and LLaVA-1.5, which reaches exactly zero group accuracy when the question comes first.

The obvious explanation is that question-first prompting damages perception, and it is wrong. Reading image patches with a logit lens, question-first makes $2.49\times$ more patches decode a content word from the question than question-last does (6.975 against 2.800 per image), and the question measurably rewrites those tokens, moving them nearly twice as far from an image-only forward pass as the system message alone. The ordering that looks most like it has found the answer is the one that gets it wrong. We call this the question-first paradox.

We locate the failure in read-out rather than perception. At the layer where the answer position attends most strongly to the question, question-first sends $2.19\times$ less attention mass there and $2.51\times$

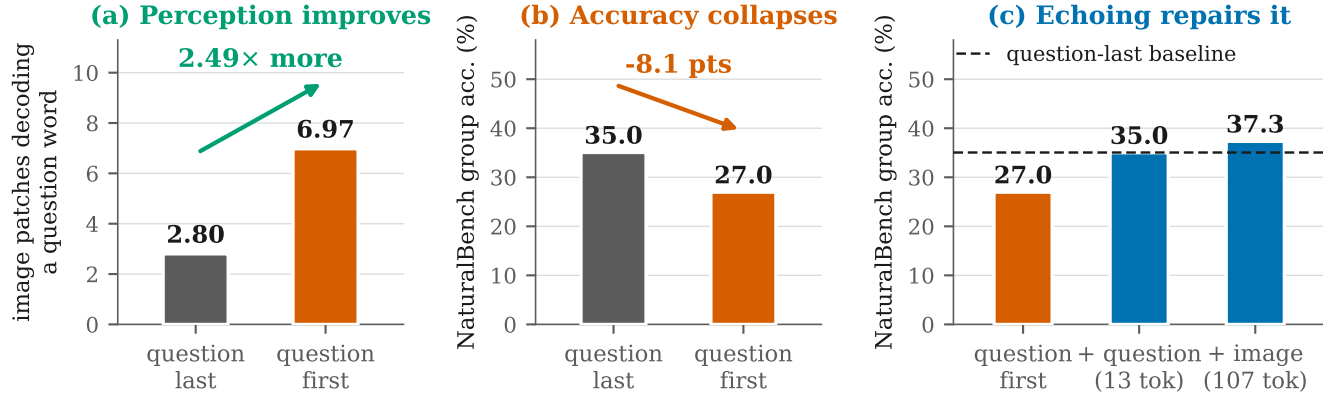


Figure 1: The question-first paradox and its repair, Qwen3-VL-8B. **(a)** Question-first makes $2.49\times$ more image patches decode a question content word (160 NaturalBench candidates), so the question reaches the image. **(b)** It still loses 8.1 points of group accuracy ($p = 1.3 \times 10^{-12}$). **(c)** Echoing the question after the image repairs it for 13 tokens; an $\frac{1}{8}$ -resolution image echo beats the baseline for 107.

more to the image. The two orderings are indistinguishable at chance through layer 25 of 36 and then diverge to $P(\text{correct})$ of 0.0986 against 0.8747. Severing attention edges shows the dependence is causal and doubly dissociated: cutting the answer-to-question path costs question-last 5.6 points and question-first nothing, while cutting the answer-to-image path costs question-first 9.6 points and question-last nothing.

That diagnosis prescribes the prompt. If the answer position cannot read the question, put something after the image that it can read. Repeating the question (STIT) costs 13 tokens, recovers 78% of the deficit, and provably cannot change perception: the image tokens stay identical to five decimals. Repeating the image (SITIT) recovers the deficit entirely and adds 1.18 points over question-last across 13 benchmarks. Neither repair supplies new information. A reversed echo works as well as a correct one, and an eighth-resolution echo matches a full one at 3% of the added tokens. A reflective prompt optimizer, given the same budget, gains only under the ordering the diagnosis says has headroom and never reaches the question-last baseline it competes with.

2 SETUP

Notation. A prompt has three sections: the system message (S), the image (I), which expands to many visual tokens, and the task or question (T). We write an ordering as the sequence of its sections, so SIT is system-image-question and STI is system-question-image. An ordering may repeat a section: STIT repeats the question after the image, SITIT repeats the image after the question, and SITIT_{rev} is SITIT with the second image copy spatially reversed. The two baselines are SIT, which is what most inference code emits by default, and STI, which is what a reader is told to do when looking for something specific.

Models. Qwen3-VL-8B is the primary model and carries every analysis. For breadth we also run Gemma-3-27B, Gemma-4-31B, Qwen2.5-VL-7B, InternVL3-8B and LLaVA-1.5. Gemma-3-27B and Gemma-4-31B are loaded in 4-bit on a single GPU. All results use greedy decoding with a fixed answer-token budget, and every ordering of a given benchmark shares one system message, one question string per item, one image preprocessing path and one scoring function, so that the ordering is the only difference between columns.

Benchmarks. The four benchmarks that carry both the accuracy ladder and the mechanistic analysis are NaturalBench (Li et al., 2024a) with 1,900 groups (7,600 image-question pairs, scored by pair, question, image and group accuracy), POPE (Li et al., 2023) with 9,000 items across its random, popular and adversarial splits, Winoground (Thrush et al., 2022) with 400 examples (1,600 items, scored by group accuracy), and RF20, a 24,449-item presence-detection benchmark built

Table 1: Ordering ladder on four benchmarks, Qwen3-VL-8B. STI (question first) is worst everywhere, SITIT (image echoing) is best or tied-best on three of four. POPE is the exception: plain question-last already wins there. *N*: NaturalBench 1,900 groups (7,600 pairs), POPE 9,000, Winoground 400 examples (1,600 items), RF20 24,449.

Benchmark	Metric	IST	SIT	STI	STIT	SITIT	SITIT_rev
NaturalBench	pair acc.	78.93	79.21	75.29	78.95	80.36	80.26
NaturalBench	group acc.	33.95	35.05	26.95	35.00	37.37	37.37
POPE	F1	88.22	88.35	86.13	87.61	88.12	88.25
Winoground	group acc.	28.00	31.75	22.25	37.50	40.25	41.00
RF20	F1	78.92	78.57	65.59	78.61	80.56	79.78

from 20 detection datasets. For the breadth study of Section 6 we add nine image benchmarks (BLINK (Fu et al., 2024), CV-Bench (Tong et al., 2024), HR-Bench, MMStar (Chen et al., 2024), MMVP, RealWorldQA, WorldMedQA-V, TallyQA (Acharya et al., 2019) and VQAv2 (Goyal et al., 2017)) and four video QA benchmarks (MVBench (Li et al., 2024b), NExT-QA (Xiao et al., 2021), MSVD-QA (Xu et al., 2017) and TGIF-QA (Jang et al., 2017)).

Significance. For NaturalBench and Winoground we pair on the group or example index and report both an exact two-sided McNemar test and a paired bootstrap 95% confidence interval over 5,000 resamples.

Interpretability tools. All mechanistic analyses use a logit lens (nostalgebraist, 2020; Belrose et al., 2023), projecting intermediate residual-stream states through the unembedding to read out per-layer top tokens, either at the answer position or over image patches. The attention knockout follows the edge-severing methodology used for factual recall in language models (Geva et al., 2023; Meng et al., 2022).

3 THE QUESTION-FIRST PARADOX

3.1 MOVING THE QUESTION EARLIER MAKES THE MODEL WORSE

Table 1 runs the six orderings on four benchmarks with Qwen3-VL-8B. Every column uses the same system message, the same question string, the same image, the same greedy decoding and the same scoring code. The only variable is where the question sits.

Question-first (STI) is the worst ordering on all four benchmarks, and the margins are not small. NaturalBench group accuracy falls from 35.05 to 26.95, Winoground from 31.75 to 22.25, and RF20 F1 from 78.57 to 65.59. On NaturalBench this is +0.081 group accuracy for question-last with a paired bootstrap 95% CI of [+0.058, +0.103] and McNemar $p = 1.3 \times 10^{-12}$; on Winoground it is +0.095, CI [+0.050, +0.140], $p = 7.7 \times 10^{-5}$.

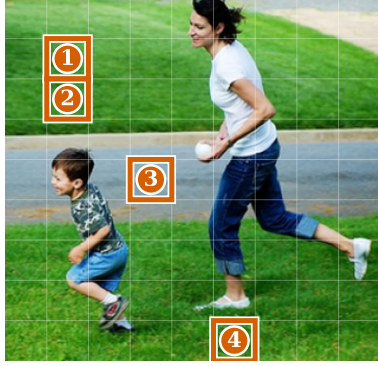
Intuition predicts the opposite. Placing the question first is what a reader would do when told to look for something specific, and it is what retrieval-style prompting recommends. The model sees identical information either way, and under causal attention the question-first layout gives every image token access to the question.

3.2 QUESTION-FIRST PERCEIVES BETTER AND ANSWERS WORSE

The obvious explanation is that question-first simply degrades the model’s perception of the image, and everything follows from that. It is wrong, and the evidence against it is the reason this paper exists.

We measure perception directly with a logit lens over image patches. For each NaturalBench candidate we count how many image patches decode, at their peak layer, a content word from the question. Across 160 candidates, question-first makes 6.975 patches per image decode a question word, against 2.800 for question-last, a factor of 2.49 and a mean gap of +4.175 patches. The effect peaks late: 76 of 160 candidates peak at layer 30 or beyond. Figure 2 shows what this looks like

“Is the adult chasing a child?” (answer: Yes)



question-last answers “Yes” (correct)		question-first answers “No” (wrong)	
1	boy	→	toddler
2	hand	→	ahead
3	throws	→	chase
4	woman	→	adult

patch words at layer 32, 46 of 81 patches change

Figure 2: The question rewrites what the image patches say, and the model still answers wrong. NaturalBench example 1007, logit-lens top-1 token per patch at layer 32 over a 9×9 grid; question-first changes 46 of the 81 patch words and four are called out. Scene vocabulary (*boy*, *hand*, *throws*, *woman*) becomes the question’s own vocabulary (*toddler*, *ahead*, *chase*, *adult*), including the answer word, yet question-first answers “No” to a question whose ground truth is Yes. Patch decodes are not localized to the object they name.

patch by patch. Asking “Is the adult chasing a child?” before the image turns *throws* into *chase*, *boy* into *toddler* and *woman* into *adult*, so the patches now decode the question’s own words, including its answer word. The model then answers “No” to a question whose ground truth is Yes, while question-last, whose patches still say *throws* and *boy*, answers “Yes”.

So the question does reach the image, and it reaches it more strongly in precisely the ordering that scores worst. The model that looks more like it found the answer is the model that gets it wrong.

Finding. Question-first is the ordering that perceives best and answers worst. It makes $2.49\times$ more image patches decode a question content word than question-last (6.975 vs 2.800 per image, $n = 160$), while scoring 8.1 points lower on NaturalBench group accuracy ($p = 1.3 \times 10^{-12}$). Degraded perception cannot explain the deficit.

3.3 THE PARADOX IS NOT ONE MODEL’S QUIRK, BUT IT IS NOT UNIVERSAL EITHER

Running the same ladder on NaturalBench for five models (Table 4 in Appendix B) shows four of the five reproduce the paradox, two of them far more severely than Qwen3-VL-8B. Qwen2.5-VL-7B loses 17.6 points of group accuracy when the question moves first ($27.74 \rightarrow 10.16$), and LLaVA-1.5 falls from 12.32 to exactly 0.00, never getting all four items of any group right. InternVL3-8B loses 7.9.

Gemma-3-27B is the exception and we flag it rather than bury it: its question-first and question-last scores are within noise (-0.006 group accuracy, $p = 0.62$). Section 6.3 shows the same for Gemma-4-31B. A model can be free of this failure. Most of the ones we tested are not.

Finding. The paradox generalizes across model families without being universal. Question-first costs Qwen2.5-VL-7B 17.6 and LLaVA-1.5 12.3 points of NaturalBench group accuracy, with LLaVA-1.5 reaching exactly 0.0, while Gemma-3-27B shows no effect ($+0.58$, $p = 0.62$).

4 ANATOMY OF THE FAILURE

Two families of explanation come to mind for Section 3, and both turn out to be wrong. The first is that question-first degrades the model’s perception of the image. The second is that it is a length or budget effect, since the two layouts sit differently in the context window. We reject the second immediately: STI and SIT contain exactly the same tokens in a different order, so their prompt

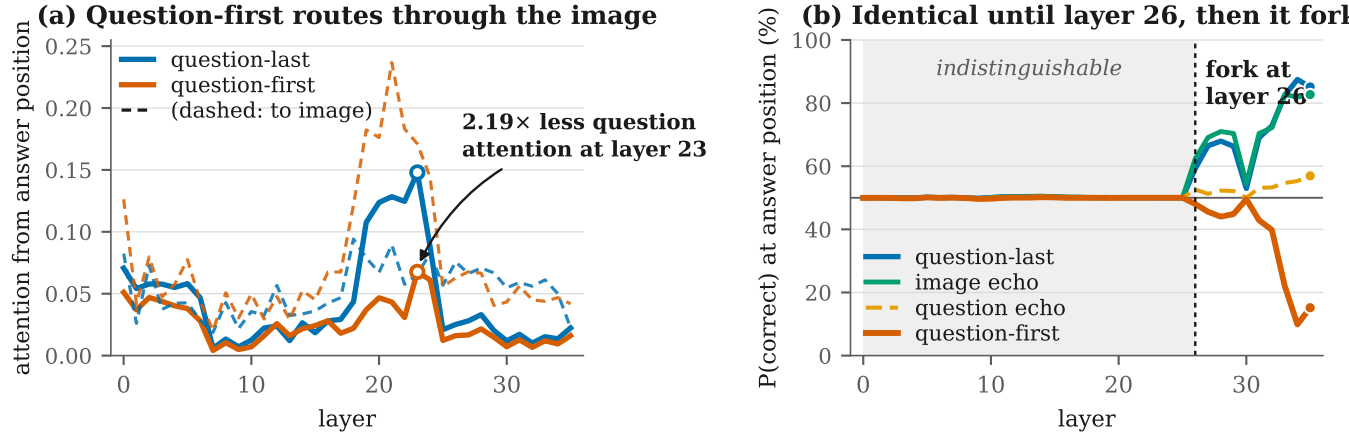


Figure 3: Where question-first breaks (150 NaturalBench pairs where it is wrong and question-last is right). (a) Attention leaving the answer position, to the question (solid) and the image (dashed): at layer 23 question-first sends $2.19\times$ less to the question. (b) $P(\text{correct})$ per layer. The two baselines sit at chance through layer 25, fork at 26 and never re-cross; both echo orderings track question-last.

lengths are identical to the token (Table 9). We spend the rest of this section rejecting the first, and locating what actually breaks.

4.1 THE QUESTION DOES REACH THE IMAGE, AND REWRITES IT

Putting the question before the image changes the image tokens themselves. We compare each ordering’s layer-18 image-token states against a run whose input is the image alone, with no system message and no question, and take the mean per-patch cosine. Under *IST*, where nothing precedes the image, the patches are the image-only patches (mean cosine 0.999998 over 7,600 pairs). Under *SIT*, where only the system message precedes them, it falls to 0.9110. Under *STI* it falls to 0.8627, with the most affected patch at 0.6149, so the question-specific rewrite is about $1.8\times$ the size of the system message’s own effect.

Causal masking gives a free control here, and it also explains a later result. Echoing the question *after* the image (*STIT*) cannot alter the image tokens, and the measurement confirms it to five decimals: $\cos(\text{STI}, \text{STIT}) = 0.99999$. Whatever *STIT* does for accuracy, it does without touching perception.

4.2 WHAT BREAKS IS READ-OUT, AND IT BREAKS LATE

If perception is intact, the failure has to be downstream of it. We probe the answer position directly on the 150 NaturalBench pairs where *STI* is wrong and *SIT* is right, tracking how much attention mass the answer position sends to the question span and to the image span at each of the model’s 36 layers.

At layer 23, where answer-to-question attention peaks for both orderings, *STI* sends 0.0676 of its mass to the question while *SIT* sends 0.1479, a $2.19\times$ deficit. Attention to the image runs the other way, peaking at 0.2367 for *STI* against 0.0942 for *SIT*. Averaged over layers 20 to 35, the ratio of question to image attention is 0.424 under *STI* and 1.065 under *SIT*. Question-first leaves the answer position staring at the image it has already encoded, and not at the question it is supposed to answer.

The timing matters as much as the size. Tracking $P(\text{correct})$ over the {Yes, No} decision at each layer on 300 pairs, the two orderings are indistinguishable at chance (≈ 0.50) through layer 25. They separate at layer 26 and never re-cross: by layer 34, *STI* is at 0.0986 and *SIT* at 0.8747. A failure that only appears in the last quarter of the network is not an encoding failure. Logit-lens $P(\text{gt})$ at the final layer tells the same story, 0.1346 for *STI* against 0.8485 for *SIT*.

Finding. The failure is read-out, not perception, and it is late. Question-first sends $2.19\times$ less attention from the answer position to the question at the peak read-out layer (0.0676 vs 0.1479), the two orderings are indistinguishable until layer 26 of 36, and they then diverge to $P(\text{correct})$ of 0.0986 versus 0.8747.

4.3 A CAUSAL TEST: SEVERING THE TWO PATHS DISSOCIATES THE ORDERINGS

The attention numbers are correlational. To show that the question path is what SIT depends on and the image path is what STI depends on, we sever each one and measure the damage. We zero the attention edges from the answer position to the question span (ko_question) or to the image span (ko_image), on 250 NaturalBench pairs, and add a control that severs an equal-length random span of pre-image text (ko_random) so that any effect of merely removing attention mass is accounted for.

The result is a clean double dissociation (Figure 4). Cutting the answer-to-question path costs SIT 5.6 points ($58.8 \rightarrow 53.2$, paired bootstrap 95% CI $[+1.99, +9.60]$) and costs STI nothing at all (+0.4). Cutting the answer-to-image path costs STI 9.6 points ($57.6 \rightarrow 48.0$) and costs SIT exactly zero. The random-span control moves both orderings by at most 0.8 points, so neither effect is an artifact of removing attention mass.

Question-last answers by reading the question. Question-first answers by reading the image, having already folded its reading of the question into those image tokens, and that is the path that does not carry the answer.

Finding. The two orderings answer through different paths, and we show it causally. Severing answer-to-question costs SIT 5.6 points (95% CI $[+1.99, +9.60]$) and STI nothing (+0.4); severing answer-to-image costs STI 9.6 points and SIT nothing (0.0); a span-size control moves both by ≤ 0.8 .

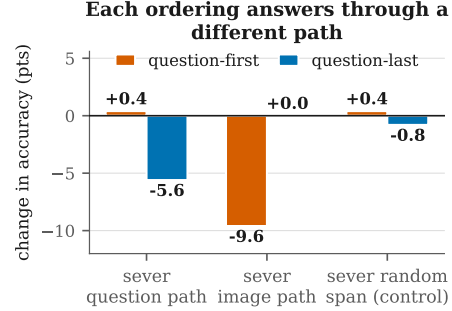


Figure 4: Attention knockout, 250 NaturalBench pairs, query side restricted to the answer position. Bars are change against the unmodified run.

5 THE DIAGNOSIS PRESCRIBES THE PROMPT

Section 4 says the answer position under STI cannot read the question, and answers through image tokens instead. The repair follows without design freedom: put something after the image that the answer position can read. There are exactly two things available, and we use both.

Question echoing (STIT). Repeat the question after the image. This restores a question span adjacent to the answer position while leaving the steered image tokens untouched, which is not merely plausible but forced: $\cos(\text{STI}, \text{STIT}) = 0.99999$ (Section 4.1), so STIT repairs read-out and provably changes nothing about perception. It costs 13 tokens.

Image echoing (SITIT). Repeat the image after the question, so the answer position has both a nearby question and image tokens encoded with the question already in context. Section 7.2 prices it.

The same construct serves two jobs, deliberately. As a diagnostic, STIT is the controlled intervention that isolates read-out: matched on content, matched on perception, differing only in whether the answer position has a question to read. As a method, it is the cheaper of the two prompts we recommend. We did not pick these orderings because they worked; they are what the mechanism leaves available.

5.1 TESTING THE PRESCRIPTION: THE ECHO’S CONTENT BARELY MATTERS

If the echoed image re-anchors read-out rather than being re-perceived, damaging its content should cost little. Reversing the second image copy (`SITIT_rev`) destroys the spatial correspondence between echoed patches and the scene while keeping token count and position identical, and it is free within noise: NaturalBench group accuracy moves from 37.37 to 37.37 ($p = 1.00$), Winoground from 40.25 to 41.00 ($p = 0.68$), POPE F1 from 88.12 to 88.25, RF20 F1 from 80.56 to 79.78. A second look that has been turned upside down works as well as a correct one. Section 7.1 makes the same point on the other axis.

Finding. The echo works by position, not by content. Reversing the echoed image copy changes nothing measurable (NaturalBench $\Delta = 0.000$, $p = 1.00$; Winoground $\Delta = +0.007$, $p = 0.68$), and an eighth-resolution echo matches a full one.

5.2 WHAT THE REPAIR IS WORTH, AND WHAT IT IS NOT

Two results keep us from overclaiming, and we state them here rather than in a closing limitations paragraph. First, question echoing alone is not reliably better than simply asking the question last: on NaturalBench `STIT` scores 35.00 against `SIT`’s 35.05 ($p = 1.00$). It is significantly better on Winoground (+0.057, $p = 0.012$) and for Gemma-3-27B on NaturalBench (+0.026, $p = 0.014$), but the honest claim is that `STIT` repairs question-first prompting rather than improving on question-last prompting.

Second, image echoing is a real gain over question-last, and this is the claim we defend: +0.085 group accuracy on Winoground for Qwen3-VL-8B ($p = 2.4 \times 10^{-5}$), +0.110 for Gemma-3-27B ($p = 8.1 \times 10^{-7}$), +0.024 on NaturalBench over `STIT` ($p = 0.016$), and +1.18 points averaged over the 13 benchmarks of Section 6. On POPE it does not help: question-last is already the best ordering there (88.35 F1 against `SITIT`’s 88.12), and we report that rather than omitting the benchmark.

6 RESULTS: THE REPAIR HOLDS ACROSS 13 BENCHMARKS AND 3 MODELS

Section 3 established the paradox on the four benchmarks where we can also run the mechanistic analysis. Here we ask whether it is a quirk of those four, and whether the repair survives contact with tasks the diagnosis was not built on. We run the full ordering ladder on 13 further benchmarks spanning single-image perception, high-resolution perception, counting, medical VQA and four video QA datasets (Table 3, Appendix A).

6.1 THE PARADOX IS LARGE, GENERAL, AND NOT A SCORING ARTIFACT

Moving the question from after the image to before it costs Qwen3-VL-8B 7.01 points on average and hurts on 12 of 13 benchmarks. The effect is far larger than the differences that usually separate published models: CV-Bench loses 18.01 points, MMStar 11.40, MMVP 9.00, NExT-QA 8.91. Nothing about the content of the prompt changed. The system message, the question wording, the image, the decoding parameters and the scoring code are identical across columns, and `STI` and `SIT` contain the same tokens in the same quantity. Only the position of the question moved. WorldMedQA-V is the single benchmark where question-first is not worse (+1.06); we report it rather than dropping it, and note it is also the smallest benchmark in the set ($N = 568$).

Finding. The question-first paradox is general, not a property of one benchmark. Across 13 benchmarks, moving the question before the image costs Qwen3-VL-8B 7.01 points on average and hurts on 12 of 13, with a worst case of 18.01 points on CV-Bench, while prompt content is held identical.

6.2 THIRTEEN TOKENS BUY BACK MOST OF THE LOSS

The two echo orderings separate the cost of the repair from its size. Echoing the question after the image (`STIT`) recovers 5.49 of the 7.01 points, or 78% of the deficit, for 13 extra tokens. Echoing the image after the question (`SITIT`) recovers all of it and then exceeds the question-last baseline by 1.18 points, winning on 9 of 13 benchmarks.

The four video QA datasets are where *SITIT* is weakest (MSVD-QA -0.78 , NExT-QA -0.26 , TGIF-QA -1.06 against *SIT*). These are the datasets where the “image” is a frame sequence, so the echo doubles an already long visual context, and TGIF-QA and MSVD-QA also have the lowest absolute accuracies in the set. We do not claim image echoing as the right choice for video.

Finding. The repair is cheap in proportion to its size. Repeating only the question after the image recovers 78% of the question-first deficit (+5.49 of 7.01 points) for 13 tokens, and repeating the image recovers all of it and adds 1.18 points over question-last, winning on 9 of 13 benchmarks.

6.3 CROSS-MODEL SCOPE: THE PARADOX IS MODEL-SPECIFIC, THE REPAIR IS NOT

Running the same ladder on Gemma-3-27B and Gemma-4-31B separates two claims that would otherwise be easy to conflate (Table 5, Appendix B). The paradox itself does not replicate on either Gemma model: question-first is on average 0.14 points *better* than question-last for Gemma-3-27B and 1.00 points better for Gemma-4-31B. Whatever makes question-first costly in Qwen3-VL-8B is not a universal property of decoder-only vision-language models, and we do not claim it is.

Image echoing survives this transfer anyway. *SITIT* is the best of the four orderings on 4 of 4 benchmarks for Gemma-4-31B and 3 of 4 for Gemma-3-27B, with mean gains over question-last of 2.28 and 1.66 points. The intervention has a wider domain than the diagnosis that motivated it, which is the honest way to state its scope: we explain why it helps Qwen3-VL-8B, and we report that it also helps two models whose read-out is not visibly broken.

Finding. The paradox and the repair have different scopes. Question-first is not harmful on Gemma-3-27B (+0.14) or Gemma-4-31B (+1.00), so the paradox is model-specific, yet image echoing is still the best ordering on 7 of 8 Gemma model-benchmark pairs (mean +1.66 and +2.28 over question-last).

7 ABLATIONS: WHAT THE ECHO IS ACTUALLY MADE OF

The diagnosis in Section 4 makes a strong claim about what the second image copy is for. It is not there to be re-perceived. It is there to put image tokens after the question so the read-out step has something local to attend to. That claim has a sharp consequence: the echoed copy should not need to be legible. We test it by degrading the echo, by pricing it, and by asking whether a prompt optimizer that can only edit text finds the same fix.

7.1 THE ECHOED COPY DOES NOT NEED TO BE LEGIBLE

We downscale only the second image occurrence in *SITIT*, leaving the first at full resolution, and sweep the echo scale over 1, $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ (Table 2). On three of four benchmarks a quarter-resolution echo matches or beats a full-resolution one: POPE moves by -0.05 F1, Winoground by $+0.75$ group accuracy, and NaturalBench by $+0.21$ pair accuracy.

Pushing to an eighth makes the point sharper. On NaturalBench, an echo carrying $\frac{1}{64}$ of the original pixels scores 37.32 group accuracy against the full echo’s 37.37, a difference of 0.001 with $p = 1.00$, while still beating question-last by $+0.023$ ($p = 0.005$). The echo can be destroyed as an image and keep its entire benefit.

RF20 is the exception, and it is the informative one. Its accuracy falls monotonically as the echo degrades (-0.50 F1 at half, -1.76 at quarter). RF20 asks whether a specific, often small object class is present in a detection-style image, so the second look is doing real perceptual work there rather than only re-anchoring. The split between RF20 and the other three says the echo has two separable jobs, and that only one of them needs pixels.

Finding. The echoed image does not need to be legible. An eighth-resolution echo, carrying $\frac{1}{64}$ of the pixels, is statistically indistinguishable from a full-resolution one on NaturalBench (37.32 vs 37.37 group accuracy, $p = 1.00$) while still beating question-last ($p = 0.005$). Only RF20, whose targets are small, needs the pixels (-1.76 F1 at quarter resolution).

Table 2: Echo-resolution sweep, Qwen3-VL-8B. Only the *second* image occurrence is downscaled; the first stays at full resolution. Three of four benchmarks are flat or better at quarter resolution. RF20, whose targets are small, degrades monotonically. Blank cells were not run.

Benchmark	Echo 1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$
POPE (F1)	88.12	88.35	88.07	
Winoground (grp)	40.25	41.75	41.00	41.75
NaturalBench (pair)	80.36	80.55	80.57	80.18
RF20 (F1)	80.56	80.06	78.80	

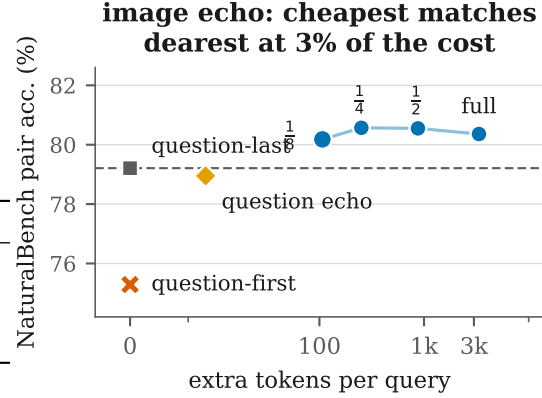


Figure 5: Cost against accuracy on NaturalBench. The x axis is extra tokens per query versus question-first (symmetric log, so 0 is shown); the dashed line is question-last. Every image echo clears that baseline, and the $\frac{1}{8}$ -resolution echo (+107 tokens) is indistinguishable from the full one (+3,352).

7.2 WHAT ECHOING COSTS

Prompt-level interventions are usually reported without their bill. We count exact token counts for each ordering with the Qwen3-VL-8B processor over 20 images spanning RF20’s constituent datasets, separating vision from text tokens (Table 9 in Appendix F).

The two echo mechanisms are three orders of magnitude apart. Repeating the question costs 13 tokens. Repeating the image at full resolution costs 3,352, which roughly doubles the prompt. Downscaling the echo collapses that cost quickly, because vision-token count falls with area: a half-resolution echo costs 875 added tokens, a quarter-resolution echo 252, and an eighth-resolution echo 107. Figure 5 puts the two axes together. The accuracy curve is flat across a $31\times$ range of echo cost, so the operating point to ship is the cheapest one.

Finding. Echo cost is a tunable knob, not a fixed doubling of the prompt. A full-resolution image echo adds 3,352 tokens per query, but a quarter-resolution echo adds 252 (7.5%) and an eighth-resolution echo adds 107 (3.2%) for indistinguishable accuracy, while question echoing adds 13.

7.3 PROMPT OPTIMIZATION FINDS PART OF THE FIX, AND ONLY WHERE THE MECHANISM SAYS IT SHOULD

A reviewer will reasonably ask whether careful prompt engineering would do the same work as echoing, making the ordering story incidental. We test this with GEPA (Agrawal et al., 2025), a reflective prompt optimizer, and use it as a prediction test rather than as a baseline to beat.

The mechanism predicts an asymmetry. GEPA may rewrite the system message text but cannot move the image, so it cannot repair a read-out failure caused by token order. It should find recoverable headroom under STI, where read-out is broken, and little or nothing under SIT, where it is not. That is what happens (Table 10, Appendix G). Under STI, GEPA improves all three benchmarks: POPE 86.41 $\rightarrow$ 87.25, VQAv2 86.11 $\rightarrow$ 88.13, NaturalBench 74.74 $\rightarrow$ 75.79. Under SIT, the same optimizer with the same budget returns the seed prompt unchanged on POPE and VQAv2, both exact ties, and regresses NaturalBench by 0.41 by overfitting its validation split.

Two further numbers matter. First, SIT’s baseline is higher than STI’s on every benchmark (88.27 vs 86.41, 88.55 vs 86.11, 78.80 vs 74.74), so the asymmetry is a headroom effect and not a quirk of the optimizer. Second, GEPA’s best STI prompt never reaches the plain SIT baseline: it closes 26% to 83% of the STI-to-SIT gap and stops there. Its optimized prompts add 26, 32 and 71 tokens per query, less than even an eighth-resolution echo, but its search costs 157,725 to 982,343 tokens

once plus a separate reflection model. Rewording the prompt recovers part of what bad ordering costs. Moving or echoing the image recovers the rest.

Finding. Prompt optimization behaves exactly as the read-out diagnosis predicts: GEPA gains under STI (POPE +0.84, VQAv2 +2.02, NaturalBench +1.04) and finds nothing under SIT (two exact ties and -0.41), because SIT has less headroom to recover. It closes 26% to 83% of the STI-to-SIT gap and never reaches the SIT baseline, so rewording is not a substitute for fixing token order.

8 DISCUSSION

The standard picture of a vision-language model answering a question is that the question selects what to look for and the model then looks. Under question-first prompting the first half of that picture is measurably correct: the question reaches the image and rewrites its tokens, and $2.49\times$ more patches decode the question’s own words. The model still answers worse. The missing step is that the evidence has to be read back out at the answer position, through attention paths that token order determines. This is why the fix is so cheap relative to its size: nothing has to be learned and nothing has to be re-perceived. The intervention is not supplying information, it is putting a read-out surface where the answer position can reach it.

Implications beyond prompt ordering. For benchmarking, an unreported prompt-layout choice is a confound of the same size as the model differences benchmarks are built to detect: question-first versus question-last moves CV-Bench by 18.01 points, so two implementations of “the same” prompt can differ by more than the gap being measured. For deployed and agentic systems, pipelines that prepend a goal to a growing context and then append retrieved images produce question-first layouts by construction and pay for it silently; appending a one-line restatement of the request after the visual content costs 13 tokens. For training recipes, the failure lives in the read-out path rather than in perception, which suggests instruction tuning that varies section order rather than fixing one canonical template, and the Gemma models show the failure is trainable away.

Limitations. The diagnosis is established on Qwen3-VL-8B. The paradox replicates on Qwen2.5-VL-7B, InternVL3-8B and LLaVA-1.5 but not on either Gemma model, so the read-out account describes a common failure mode rather than a property of the architecture; image echoing still helps the Gemma models, but the mechanism we document cannot be the reason there. Image echoing is also not uniformly right: it does not beat question-last on POPE (88.12 against 88.35 F1), and on the four video QA benchmarks it loses to question-last on three of four. Finally, our knockout result depends on restricting the query side to the answer position; widening it does not reproduce the question-side half of the dissociation, and the span-size control is itself large there, so we treat the broader scope as uninformative rather than as contrary evidence (Appendix I).

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A FULL ORDERING LADDER ACROSS 13 BENCHMARKS

B CROSS-MODEL LADDERS IN FULL

C SIGNIFICANCE TESTS IN FULL

Paired on group index (NaturalBench) or example index (Winoground), exact two-sided McNemar plus a paired bootstrap 95% CI over 5,000 resamples, seed 0. Accuracies are group accuracy.

Table 3: Full ordering ladder for Qwen3-VL-8B across 13 benchmarks (video QA below the rule). Best of the four orderings in bold. The echo column gives SITIT with the second image copy at half resolution. Question-first costs 7.01 points on average; image echoing recovers all of it and adds 1.18.

Benchmark	N	Ordering				Echo $\frac{1}{2}$ SITIT	Δ vs SIT	
		SIT	STI	STIT	SITIT		STI	SITIT
BLINK	793	53.09	44.51	52.21	53.97	50.44	-8.58	+0.88
CV-Bench	2,638	81.35	63.34	79.98	83.55	83.70	-18.01	+2.20
HR-Bench	1,600	67.81	59.25	63.94	69.75	68.44	-8.56	+1.94
MMStar	1,500	53.33	41.93	51.47	55.13	56.07	-11.40	+1.80
MMVP	300	73.00	64.00	72.33	77.67	76.00	-9.00	+4.67
RealWorldQA	765	69.28	63.92	67.19	70.33	70.33	-5.36	+1.05
WorldMedQA-V	568	53.35	54.40	56.87	57.22	57.57	+1.06	+3.87
TallyQA	38,589	79.14	74.15	77.85	79.99	79.36	-5.00	+0.84
VQAv2	214,354	81.66	78.72	80.49	81.30	81.38	-2.94	-0.36
MVBench	2,600	64.08	55.88	62.23	64.62	66.08	-8.19	+0.54
NExT-QA	8,564	78.21	69.30	76.05	77.95	78.11	-8.91	-0.26
MSVD-QA	13,157	40.05	35.52	36.61	39.27	39.51	-4.53	-0.78
TGIF-QA	25,751	36.32	34.59	33.62	35.26	35.50	-1.74	-1.06
<i>Mean over 13</i>							-7.01	+1.18

Table 4: NaturalBench ladder across five models (1,900 groups). Four of five models lose accuracy when the question moves before the image. Missing entries were not run. LLaVA-1.5’s question-first group accuracy is exactly zero.

Model	Metric	SIT	STI	STIT	SITIT	Δ (STI-SIT)
Qwen3-VL-8B	pair acc.	79.21	75.29	78.95	80.36	-3.92
	group acc.	35.05	26.95	35.00	37.37	-8.10
Qwen2.5-VL-7B	pair acc.	76.05	64.95	71.83	75.71	-11.10
	group acc.	27.74	10.16	19.11	27.63	-17.58
InternVL3-8B	pair acc.	79.63	75.61	79.76	80.80	-4.02
	group acc.	35.53	27.63	36.47	39.32	-7.90
LLaVA-1.5	pair acc.	66.87	50.26	66.29	n/a	-16.61
	group acc.	12.32	0.00	12.05	n/a	-12.32
Gemma-3-27B	pair acc.	73.34	73.21	74.64	74.61	-0.13
	group acc.	22.63	23.21	25.26	25.53	+0.58

D READ-OUT PROBE AND DECISION LAYER IN FULL

Qwen3-VL-8B, 36 layers. The disagreement set is the 150 NaturalBench pairs where STI is wrong and SIT is right; the decision-layer analysis uses 300 pairs. a_q and a_{img} are attention mass from the answer position onto the question and image spans, averaged over heads.

$P(\text{correct})$ over the {Yes, No} decision, layers 26 to 35, on the 300-pair disagreement set: STI 0.480, 0.456, 0.440, 0.449, 0.495, 0.429, 0.398, 0.222, 0.099, 0.152; SIT 0.593, 0.665, 0.680, 0.663, 0.529, 0.691, 0.728, 0.824, 0.875, 0.851; SITIT 0.624, 0.691, 0.710, 0.704, 0.546, 0.704, 0.722, 0.827, 0.819, 0.827; STIT 0.526, 0.513, 0.523, 0.521, 0.500, 0.532, 0.533, 0.547, 0.553, 0.570. Both orderings sit at ≈ 0.50 through layer 25.

E PER-PATCH COSINE TO AN IMAGE-ONLY FORWARD PASS

Layer 18, images resized to 448×448 , $14 \times 14 = 196$ patches, Qwen3-VL-8B. Mean per-patch cosine between each ordering’s image-token hidden states and a forward pass on the image alone.

Sanity checks: $\cos(\text{base}, \text{IST}) = 0.999997$ and $\cos(\text{STI}, \text{STIT}) = 0.999992$ at $n = 7,600$.

Table 5: Cross-model check on the four benchmarks run for all three models. The question-first paradox does not replicate on either Gemma model, yet image echoing is still the best ordering on 7 of 8 model-benchmark pairs.

Model	Benchmark	N	Ordering				Δ vs SIT	
			SIT	STI	STIT	SITIT	STI	SITIT
Gemma-3-27B	BLINK	793	44.51	43.51	42.75	44.89	-1.01	+0.38
	MMVP	300	72.67	72.33	72.00	74.67	-0.33	+2.00
	TallyQA	38,589	63.00	67.60	68.93	69.97	+4.60	+6.96
	VQAv2	214,354	64.77	62.07	63.04	62.07	-2.70	-2.70
	Mean						+0.14	+1.66
Gemma-4-31B	BLINK	793	54.48	56.12	56.12	56.49	+1.64	+2.02
	MMVP	300	84.67	86.00	86.67	87.33	+1.33	+2.67
	TallyQA	2,000	80.65	80.50	81.20	82.60	-0.15	+1.95
	VQAv2	3,000	73.50	74.66	75.65	76.00	+1.16	+2.50
	Mean						+1.00	+2.28

Table 6: Significance for consecutive rungs of the ladder and for the headline comparisons. STI versus SIT is highly significant for Qwen3-VL-8B on both benchmarks and absent for Gemma-3-27B. SITIT versus SIT is significant on Winoground for both models.

Benchmark	Model	Comparison	acc A	acc B	Δ (95% CI)	McNemar p
NaturalBench	Qwen3-VL-8B	STI vs SIT	0.269	0.351	+0.081 [+0.058, +0.103]	1.3×10^{-12}
		SIT vs STIT	0.351	0.350	-0.001 [-0.022, +0.020]	1.00
		STIT vs SITIT	0.350	0.374	+0.024 [+0.005, +0.043]	1.6×10^{-2}
		SITIT vs SITIT_rev	0.374	0.374	+0.000 [-0.010, +0.011]	1.00
	Gemma-3-27B	STI vs SIT	0.232	0.226	-0.006 [-0.026, +0.015]	0.62
		SIT vs STIT	0.226	0.253	+0.026 [+0.006, +0.046]	1.4×10^{-2}
		STIT vs SITIT	0.253	0.255	+0.003 [-0.015, +0.021]	0.82
	Winoground	Qwen3-VL-8B	STI vs SIT	0.223	0.318	+0.095 [+0.050, +0.140]
SIT vs STIT			0.318	0.375	+0.057 [+0.015, +0.102]	1.2×10^{-2}
SIT vs SITIT			0.318	0.403	+0.085 [+0.045, +0.125]	2.4×10^{-5}
STI vs SITIT_rev			0.223	0.410	+0.187 [+0.140, +0.235]	1.2×10^{-13}
Gemma-3-27B		STI vs SIT	0.212	0.207	-0.005 [-0.052, +0.043]	0.92
		SIT vs STIT	0.207	0.253	+0.045 [+0.000, +0.087]	6.3×10^{-2}
		SIT vs SITIT	0.207	0.318	+0.110 [+0.070, +0.152]	8.1×10^{-7}

F EXACT TOKEN COST PER ORDERING

G GEPA CONFIGURATION AND TRAINING COST

GEPA (Agrawal et al., 2025) optimizes the system message only. The task model is Qwen3-VL-8B; the reflection model is Gemma-3-27B on a separate GPU, since reusing the task model as its own reflector produced proposals that never passed GEPA’s strict-improvement acceptance test. Training and validation examples are disjoint from the held-out evaluation set.

H LOGIT-LENS WORD DIVERGENCE BETWEEN STI AND IST

To ask whether question-first and image-first prompting diverge in the words they are internally predicting, we decode the top-1 token at every (layer, generation step) position under both orderings on 100 samples, discard positions whose two words are identical or WordNet synonyms, and discard

Table 7: Read-out probe on the disagreement set. STI peaks lower on the question and much higher on the image, and the mid-to-late band ratio inverts.

Ordering	probe acc	max a_q (layer)	max a_{img} (layer)	$a_q:a_{img}$ @L24	final $P(gt)$
STI	0.073	0.0676 (23)	0.2367 (21)	0.424	0.1346
SIT	0.887	0.1479 (23)	0.0942 (18)	1.065	0.8485
STIT	0.600	0.1437 (23)	0.1448 (21)	0.986	0.5912
SITIT	0.847	0.1286 (23)	0.0647 (26)	1.544	0.8374
SITIT_rev	0.867	0.1218 (23)	0.0617 (26)	1.624	0.8503

Table 8: The question rewrites the image tokens. IST is the image-only baseline by construction; STIT is identical to STI because causal masking prevents a later question copy from affecting earlier patches.

Ordering	n	mean cos	median	min
IST	7,600	0.999998	1.000000	0.999919
SIT	30	0.911017	0.921399	0.722478
STI	7,600	0.862723	0.873233	0.614870
STIT	7,600	0.862727	0.873237	0.614835

any position where either side is not a dictionary word, since intermediate layers frequently project onto subword fragments.

Genuine word changes are rare and concentrated: 8 across 100 samples, of which 6 occur at layer 16, one at layer 8 and one at layer 20, with none at layers 0, 4, 12, 24 or 27. Divergence at the level of decoded output words is therefore sparse and mid-network, consistent with the read-out account in Section 4.2, where the two orderings are indistinguishable at the decision level until late and differ mainly in routing rather than in vocabulary.

I REPRODUCTION NOTES

Every accuracy in this paper is read from a per-run result file that records the model, the ordering, the system prompt, the decoding budget and the sample count. Two details are worth flagging for anyone re-running the analysis.

The attention knockout must restrict the query side to the answer position. With the query side widened to every position after the question, the question-side half of the dissociation does not reproduce (severing answer-to-question raises SIT by 2.8 points, CI $[-0.084, +0.028]$), and the random-span control is itself worth 10 points on STI, so that configuration is uninformative rather than contradictory.

The Gemma-3-27B RF20 run is degenerate and we exclude it from all RF20 claims. It answers “yes” to roughly 96% of items under every ordering (yes_ratio 0.955 to 0.964, recall 0.979 to 0.991), so its accuracy sits at the positive base rate and the spread across orderings is 0.003.

Table 9: Exact token cost per query, mean over 20 images spanning RF20’s constituent datasets, Qwen3-VL-8B. Δ is against the STI baseline. STI and SIT contain the same tokens in a different order, so their totals are identical. Repeating the question is nearly free; repeating the image is not, but its cost falls with echo area because vision tokens scale with pixels.

Prompt	Total	Image	Text	Δ total	Δ image	Δ text
STI (baseline)	3385.8	3335.8	50.0	n/a	n/a	n/a
STIT (question echo)	3398.8	3335.8	63.0	+13.0	+0.0	+13.0
SITIT, echo 1	6737.5	6671.5	66.0	+3351.8	+3335.8	+16.0
SITIT, echo $\frac{1}{2}$	4260.7	4194.7	66.0	+874.9	+858.9	+16.0
SITIT, echo $\frac{1}{4}$	3637.6	3571.6	66.0	+251.8	+235.8	+16.0
SITIT, echo $\frac{1}{8}$	3492.6	3426.6	66.0	+106.8	+90.8	+16.0

Table 10: GEPA prompt optimization under two orderings, Qwen3-VL-8B, on held-out evaluation sets. GEPA finds real gains under STI and nothing under SIT, as the read-out diagnosis predicts. It never reaches the SIT baseline it is competing with.

Benchmark	Order	N	Accuracy		Δ	Gap closed
			Seed	GEPA		
POPE	STI	8,800	86.41	87.25	+0.84	45%
	SIT	8,800	88.27	88.27	+0.00	n/a
VQAv2	STI	10,000	86.11	88.13	+2.02	83%
	SIT	10,000	88.55	88.55	+0.00	n/a
NaturalBench	STI	7,000	74.74	75.79	+1.04	26%
	SIT	7,000	78.80	78.39	-0.41	n/a

Table 11: GEPA search budget and cost, STI ordering. Training tokens are a one-time cost; the per-query cost is the prompt-length delta in Table 12.

Benchmark	train	val	metric calls	training tokens	wall clock (s)
POPE	100	60	400	157,725	124.3
VQAv2	150	80	1,008	396,253	232.4
NaturalBench	250	120	1,540	982,343	528.7

Table 12: Prompt-length cost of GEPA’s optimized system messages, in tokens per query. Under SIT on POPE and VQAv2 the optimizer returned the seed prompt, so the delta is zero.

Benchmark	Order	seed	optimized	Δ
POPE	STI	315	341	+26
	SIT	316	316	+0
VQAv2	STI	313	345	+32
	SIT	314	314	+0
NaturalBench	STI	216	287	+71
	SIT	217	381	+164